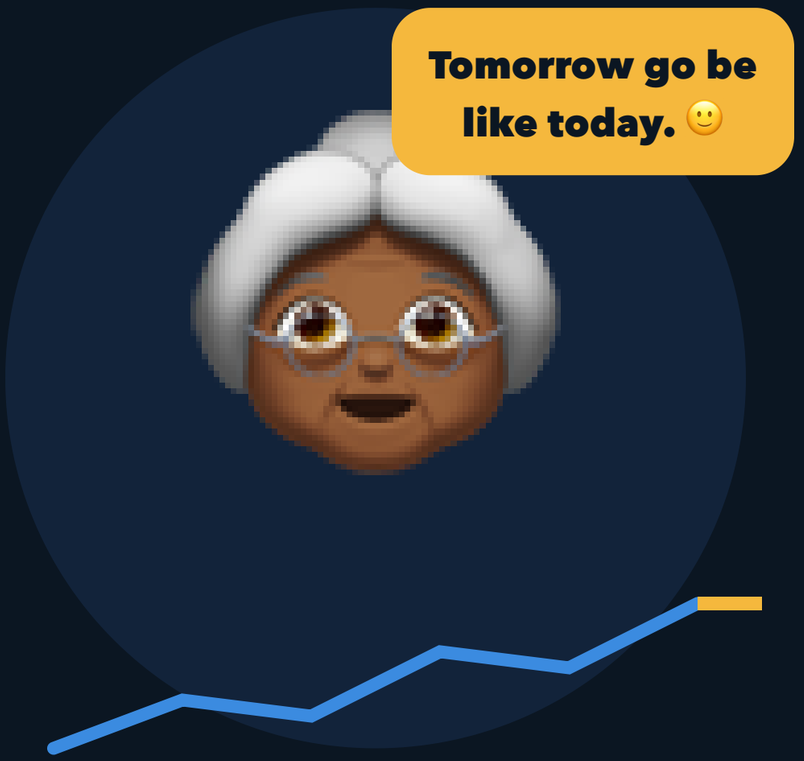
WEEK 5 · DAY 2

Forecasting Baselines

Grandma's forecast: “tomorrow will be like today.”
She's harder to beat than you think. 👵🏾



TODAY'S MISSION · 2 HOURS

Beat Grandma. Then set the bar.



Meet the series

trend · season · noise



Four baselines

the cheapest forecasts



Score them

MAE, in tickets

Divide by size

the trick that wins

⏱️ 10 min hook · 🎤 30 min slides · 👩🏾‍💻 60 min hands-on · 🏁 20 min wrap-up

QUICK QUESTION

Who forecasts next week's NOC tickets? 🧾



Grandma

"How many tickets tomorrow? The same as today, my child." No laptop. No model. Fifty years of being right.

free · one line of code



The consultant

Big slides, a machine-learning model, a ₦25 million invoice. "Our AI forecasts tickets."

expensive · never compared

If the AI can't beat Grandma, the invoice is for nothing.
Grandma's forecast has a name: a **baseline**. 👵🏾

A NEW QUESTION FOR THE LAST TWO WEEKS

From “is it wrong now?” to “what comes next?”

🚨

Weeks 3–5: detection

Look at a reading that already happened. Is it normal? Control charts, CUSUM, the forest.

🔮

Weeks 5–6: forecasting

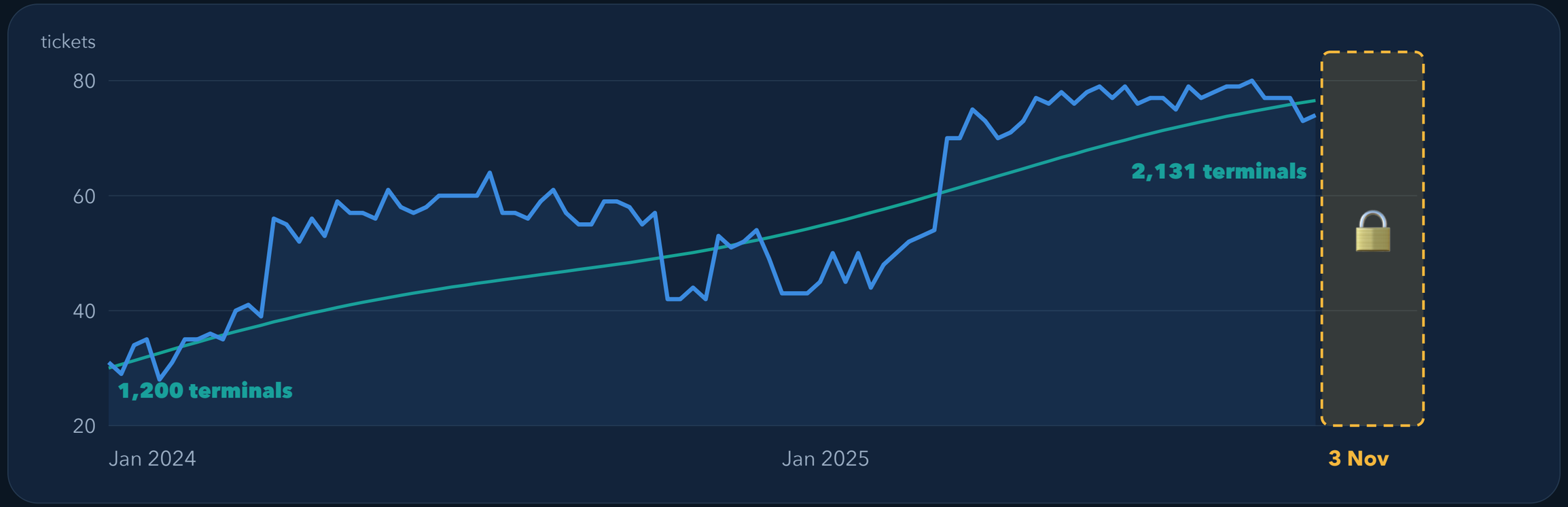
Guess a reading that **hasn't happened yet**. How many tickets next week? When will the battery reach its limit?

A forecast is a guess about the future made **only from the past**.

If your forecast secretly used next week's number, it's not a forecast. It's expo. 🕵🏾

OPS_TICKETS_WEEKLY · WEEK 6 DATA FOLDER

NOC tickets, every week for two years



104 weeks · 1 Jan 2024 – 22 Dec 2025

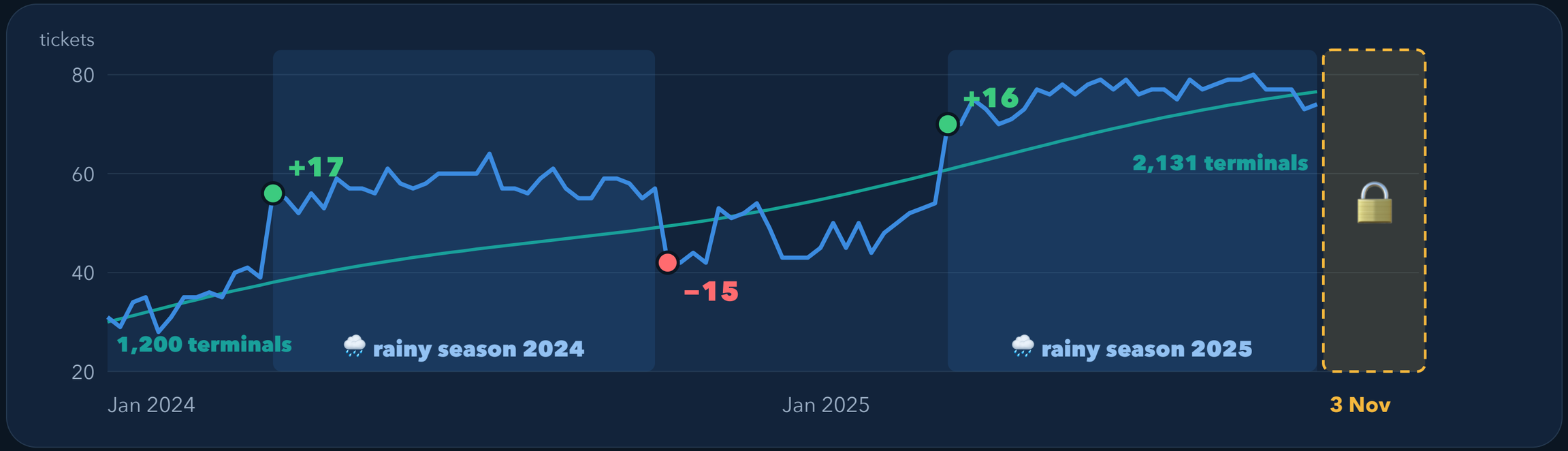
0 gaps

28–80 tickets a week

1,200 → 2,181 terminals

WHAT'S INSIDE THE SERIES

Trend + a season that comes in steps + noise



📈 Trend

The network grows every week, and tickets grow with it.

🌧️ Season

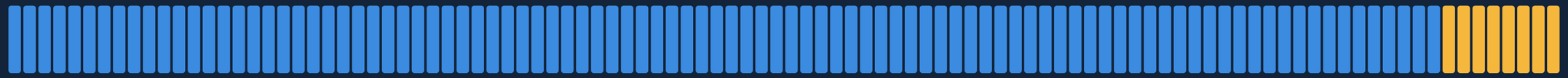
Rain arrives in April: a jump. It leaves in November: a drop. December adds a bump.

🎲 Noise

Week-to-week wobble no forecast should chase.

THE RULE FROM WEEK 5 · DAY 1

Lock the last 8 weeks away 🔒



🏋 **96 training weeks · 1 Jan 2024 - 27 Oct 2025**

🔒 **8 test weeks**

```
H = 8                                   # forecast 8 weeks ahead
train, actual = y.iloc[:-H], y.iloc[-H:]    # 96 weeks to learn from, 8 to be judged on
assert train.index.max() < actual.index.min()
```

Every forecast today is made on **27 Oct 2025** and must guess all 8 weeks **blind**. No expo.

PART 1

Four baselines 

The cheapest forecasts there are. Every one is a real forecast.

CHEAPEST FIRST · EACH MAKES ONE ASSUMPTION

Four ways Grandma could think



Naive

"Same as last week."

```
series.iloc[-1]
```

74

every week



Seasonal naive

"Same week last year."

```
series.iloc[-52:]
```

42–54

copied from Nov–Dec 2024



Moving average

"The average of the last 8 weeks."

```
series.iloc[-8:].mean()
```

77

every week



Exp. smoothing

"A running level: each week pulls it 30% of the way."

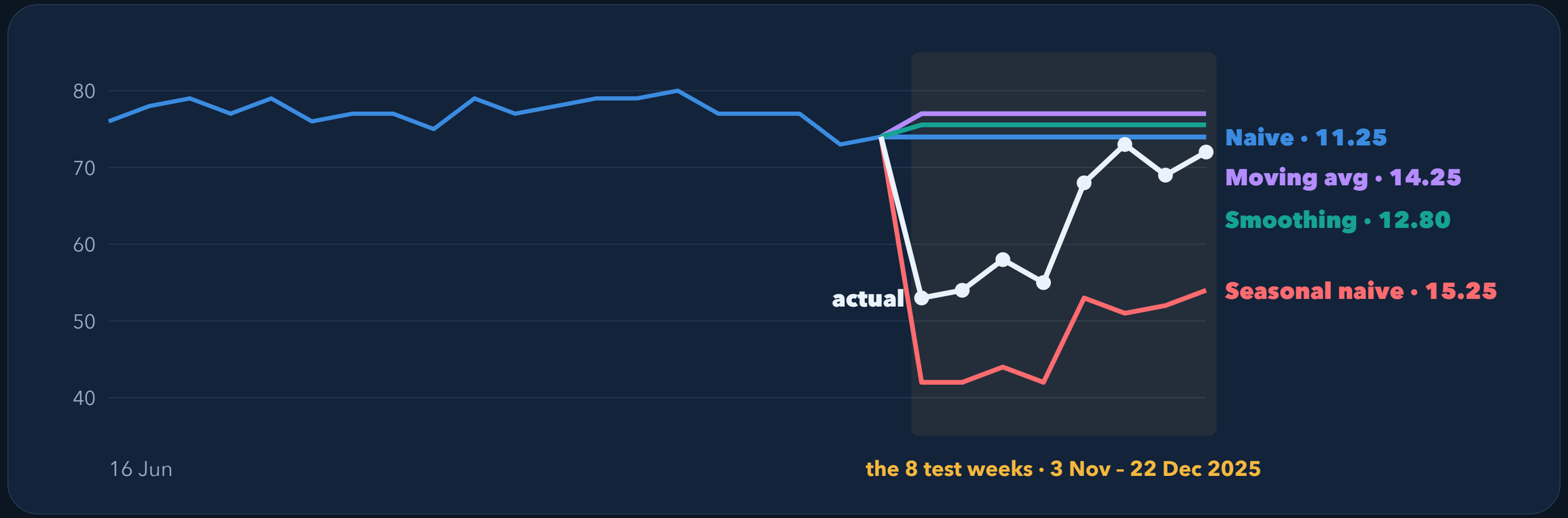
```
a*v + (1-a)*level
```

75.6

every week (α = 0.3)

UNLOCK THE 8 WEEKS 🔒

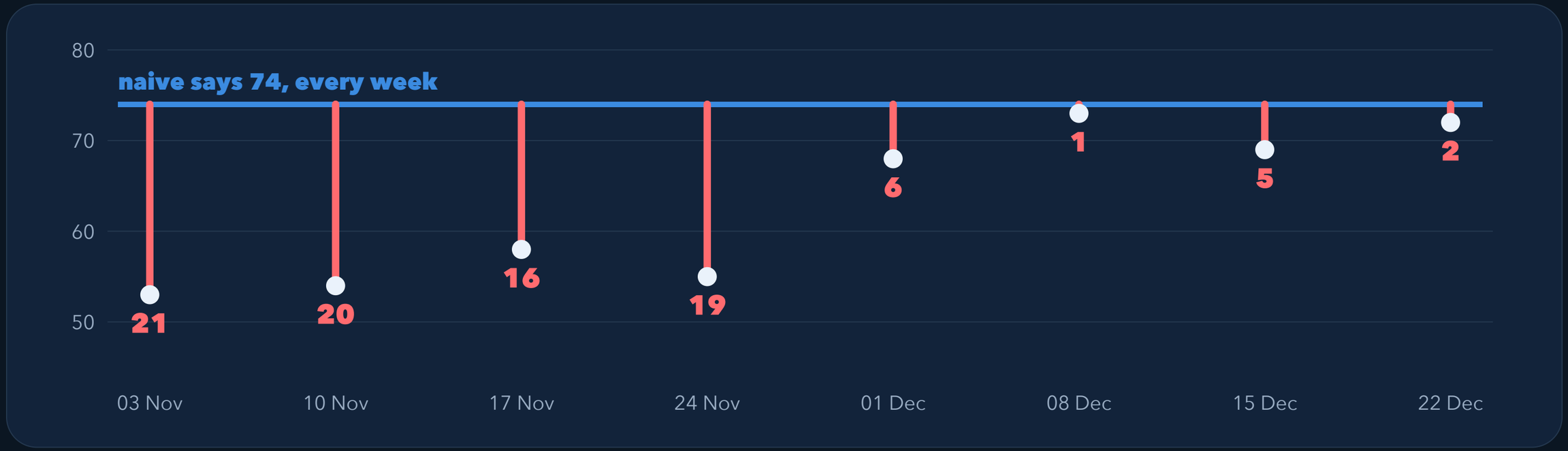
Four forecasts vs what really happened



White = actual. Naive and smoothing sail over November; seasonal naive sits **far below** all 8 weeks.

HOW WRONG, IN PLAIN WORDS?

MAE: the typical miss, in tickets

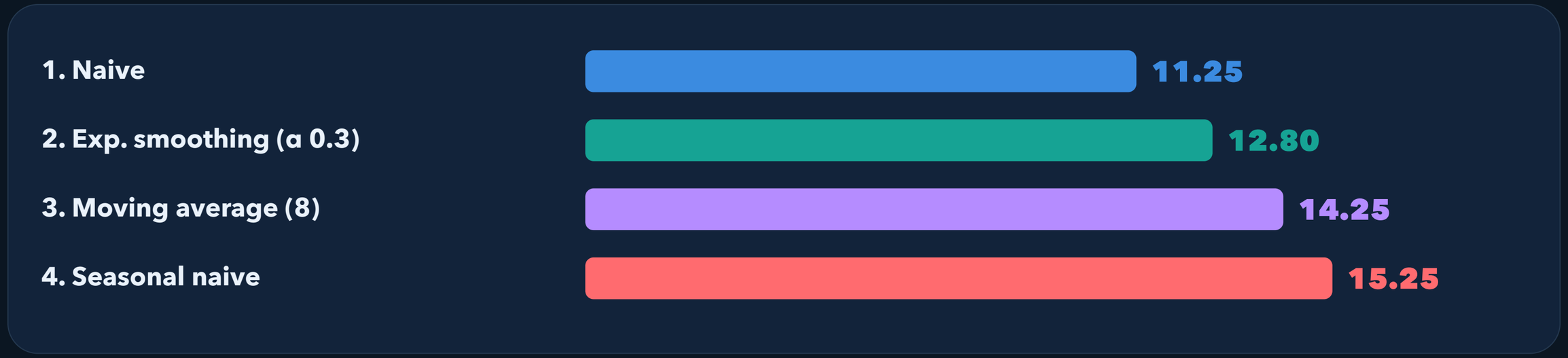



```
(21 + 20 + 16 + 19 + 6 + 1 + 5 + 2) ÷ 8 = 11.25
```

"On a typical week, naive was off by about **11 tickets**."

SAME 8 WEEKS · SAME SCORER · LOWER IS BETTER

The scoreboard 🏁



Grandma's naive wins: **11.25**. Seasonal naive comes **last**, on a seasonal series. 🤔

WHY DID SEASONAL NAIVE COME LAST?

It copied a smaller network 📡

a year before the forecast (28 Oct 2024)

1,581

terminals

Nov–Dec 2024: 42–54 tickets a week

the forecast origin (27 Oct 2025)

2,131

terminals: **+35%**

same weeks, a much bigger network

The season was right. The **size** was wrong. It missed by 15.25 tickets a week, every week too low.

AND WHY DID NAIVE “WIN”?

It copied the wrong season 🌧️→☀️

last training week · 27 Oct 2025 · rainy season

74

tickets

—

first test week · 3 Nov 2025 · dry season

53

tickets

Naive was too high **every** week: off by 16–21 tickets all November.
It won because the others were **even worse**, not because it was good.

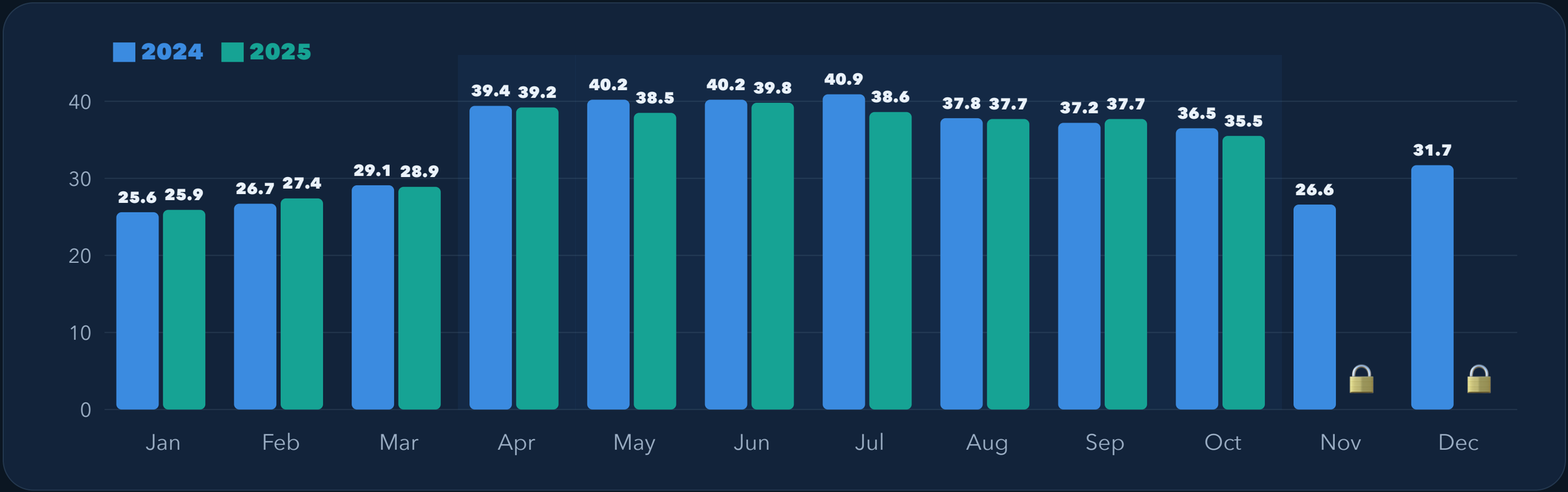
PART 2

Divide by the size ÷

The trick the original code book never tries.

TICKETS PER 1,000 TERMINALS · TRAINING WEEKS ONLY

Divide by the network: the season repeats



January 25.6 then 25.9. April 39.4 then 39.2. Never more than **2.3** apart. The rainy season lifts the rate by **37.5%**.

GRANDMA, ADJUSTED FOR THE BIGGER FAMILY 👵🏾+

Rate seasonal naive



Last year's rate

42 tickets ÷ 1,588 terminals

26.4 per 1,000

×



Today's network

at the forecast origin

2,131 terminals

=



Forecast

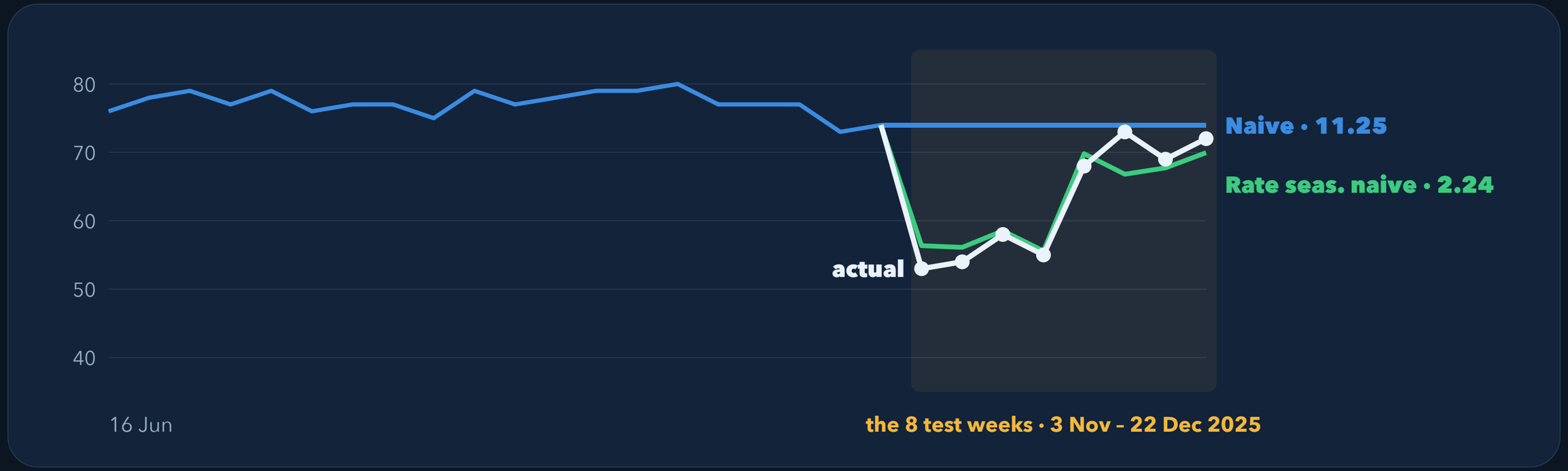
week of 3 Nov 2025

56.4

```
rate = series / terminals                                  # per terminal
return rate.iloc[-52:].values[:h] * float(terminals.iloc[-1])   # x today's size
```

SAME 8 WEEKS · SAME SCORER · STILL BLIND

It follows the drop and the growth 🎯



2.24 tickets a week, against naive's 11.25.
5× better, and it forecast all 8 weeks on 27 Oct.

YOU'VE DONE THIS BEFORE

Divide out what you already know



Week 4 · Day 3

Amplifier temperature ÷ **RF drive**. Hot because it works hard is normal; hot for its drive is EV04.

`temp_per_drive`



Today

Tickets ÷ **terminals**. More tickets because the network grew is normal; the rate shows the season.

`tickets_per_terminal`

NEPA bill per room, fuel per kilometre, jollof per guest: **a rate travels, a count doesn't**. 🍛

THE BAR FOR EVERY LATER MODEL

Every fancier model must beat this 🪜



Not the baseline its name suggests. Not the one that's easy to beat. **The best one.**

BEFORE YOU CELEBRATE…

One split can lie 🎲



The flat training average

Average all 96 training weeks (57.9) and say it every week. The forecast the code book calls "the wrong baseline".

7.77

MAE on these 8 weeks

It "beat" all four code-book baselines here. 😳

The dry-season drop and the network growth happened to cancel out on these 8 weeks.

Skill or luck? **Week 6 • Day 1** tests every method on dozens of different 8-week stretches.

YOUR TURN · 60 MINUTES

Beat Grandma yourself 👵🏾

1. Open **W5D2-exercise**, save a copy, run STEP 0
2. **A** Load the weeks, lock the last 8 away
3. **B** Write the four baselines, score them
4. **C** Divide by the size: the rate forecast
5. **D** The flat average's lucky score



14 checks to pass

🌶️ Pepper task: change α and see the expo trap

Keep the **rule book** open next to you.

TODAY IN 3 LINES

What you now know



Baselines first: naive, seasonal naive, moving average, smoothing. Cheap, and real forecasts.



Score honestly: lock the last weeks away, forecast blind, MAE in tickets, same weeks for all.



Divide by the size: last year's rate × today's network = 2.24. That's the bar.

Next 👉 **Regression forecasts + honest ranges**: “I'm on my way” vs “30 to 90 minutes”. 🚗

Final project: bring your temporal split + baseline comparison table to Checkpoint 4 (next session).